

Deep Domain Adaptation for Non-Intrusive Load Monitoring Based on a Knowledge Transfer Learning Network

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Abstract—Understanding customers' energy consumption at the individual appliances level is crucial for the planning and implementation of demand response (DR) programs. The appliances' usage profiles can be disaggregated from whole-house energy consumption data using non-intrusive load monitoring (NILM) methods. The appliance load patterns of each customer are considerably different, which make it challenging to train a model with strong generalization ability. In this paper, a novel methodology using transfer knowledge between domains for NILM is proposed. A temporal convolutional network is developed to learn the dynamic features of individual appliance load. A domain adaption loss is used to quantify the domain distribution discrepancy between source and target domain representation. By jointly optimizing domain adaptation and energy disaggregation, an invariant representation across domains for the individual appliance states can be learned. Data experiments on ground truth data validate the accuracy and the robustness of the proposed model, and demonstrate its superior transferability and application potential under those scenarios of data shortage.

Index Terms—Non-intrusive load monitoring, energy disaggregation, transfer learning, domain adaptation, temporal convolutional network.

NOMENCLATURE

Acronyms

EE	Energy efficiency
DR	Demand response
NILM	Non-intrusive load monitoring
CNN	Convolutional neural network
LSTM	Long short-term neural network
HMM	Hidden Markov model
MMD	Maximum mean discrepancy
TLN	Transfer learning network
TCN	Temporal convolutional network
MSE	Mean square error.

Sets and Indices

t	Index of time
l	Index of layer
p, q	Index of TCN block
x	Set of whole-house load
y^i	Set of i -th appliance usage
$\mathcal{D}_S$	Set of source domain data
$\mathcal{D}_T$	Set of target domain data
X_S	Set of whole-house load of source domain
Y_S	Set of appliance usage of source domain
X_T	Set of whole-house load of target domain
Θ	Set of TLN parameters.

Parameters

d	Dilation factor
k	Filter size
μ	Weighting factor of two distribution distances
λ	Weighting factor of the regression loss and domain adaptation loss
T	Length of the sliding window
l_s	Stride length of the sliding window
n_{tcn}	Number of shared TCN layers
n_{fc}	Number of fully connected layers.

Variables

ε_t	Measurement noises at time t
n_s	Number of source data
n_t	Number of target data
x_p	Input of the p -th block in a shallower location
x_q	Output of the q -th block in a deeper location
Z_S^l	Feature representation of l -th layer for source data
Z_T^l	Feature representation of l -th layer for target data.

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I. INTRODUCTION

OPTIMIZED energy management by understanding the appliances energy usage in buildings has raised great interest in the past decade, since residential and commercial electricity consumption represents approximately 60% of the global energy consumption [1]. In order to improve the energy efficiency and reduce greenhouse emissions for environmental sustainability, it is essential to provide accurate and fine-grained power usage information and operation patterns of individual appliances, which can facilitate the interaction between the end-users and energy providers.

An easy and straightforward method to obtain appliances usage profile is to install submeters for each appliance within a household and record the instant power readings. However, this intrusive load monitoring approach is not feasible because of the high expenses of equipment and maintenance difficulties. In order to avoid using these submeters, an alternative way is thus firstly proposed by Hart which is called non-intrusive load monitoring (NILM) [2] that disaggregates the mixture whole-house energy consumption into appliance level as an instance of single-channel blind-source separation problem [3]. The wide application of smart meters has enabled large amounts of energy consumption data to become available for NILM.

Because of the diverse operations of multiple individual appliances and uncertain customers' energy consumption preferences, NILM is an inherently challenging problem. Recently, some data-driven approaches have been developed to tackle this problem, mainly including supervised learning methods, unsupervised learning methods and semi-supervised learning methods. Supervised NILM methods need both the whole-house data and the individual appliance consumption data for model training. In [4], [5], a sequence-to-sequence convolutional neural network (CNN) is established to decompose a given window of the whole-house load into a sequence of target appliance values in the same length. Different from that, [6] uses a sequence-to-point architecture, where the input is a window of the aggregate power and the output is a single-midpoint of the individual appliance power. Reference [7] applies a long-short term memory network (LSTM) to extract dominant appliance usage from the aggregate power signals which are collected in a low sampling rate. These supervised methods are trained and tested on the same dataset and thus the performance is unsatisfactory and not guaranteed on any other unseen houses and unseen appliances. Besides, it is expensive and time-consuming to collect enough labeled data, which limits its use in practice. By contrast, unsupervised NILM methods do not require labeled data for pre-training. A typical method is the hidden Markov model (HMM), which can reveal the internal states of individual appliances. In [8], an HMM model is applied for capturing both dynamics and magnitudes of energy consumption. Reference [9] uses factorial HMM (FHMM) to separate the space cooling usage from residential smart meter data. However, for these HMM-based approaches, the computational complexity of the solution spaces grows rapidly as the amount of power consumption states increases [10]. As a trade-off between supervised and unsupervised learning methods, the semi-supervised approaches that learn from a small number of labeled observations and a much larger unlabeled dataset are much suitable for NILM application. A discrete wavelet transforms of power signal method is applied in [11] to learn a time-scale representation of load pattern. Although some high frequency transient features can be extracted, there still exists much difference between appliances and domains. As a result, the generalization ability of these learning methods using wavelet transforms is poor. Reference [12] proposes a mean teacher network only using low labeling rate data of a test customer. However, this trained model cannot be extended to other customers.

Basically, there are three issues that need to be improved in these semi-supervised approaches.

1) A major hypothesis of semi-supervised approaches is that the training data and the testing data should have similar data distributions. However, in practice, this hypothesis may not hold for NILM problems due to the differences between appliances with different working patterns and customers' usage habits. For practical applications of NILM, we need to apply the model trained on the labeled dataset to the testing buildings which are not present in the training dataset and only the unlabeled data is available. The data distributions between the training samples (the source domain) and the testing buildings (the target domain) are not exactly the same, and this domain discrepancy may lead to a decrease in the model performance of trained model on the unseen buildings [13].

2) Since the individual appliances' energy consumption may be a function of time, devices, outside temperatures or other dynamic factors, the labeled data obtained in one time period may not follow the same distribution in another time period. The temporal variations of appliances' energy consumption pattern have not been precisely represented.

3) Even if some broad covering datasets are available to build the load disaggregation model, extracting suitable features such as rising and falling edges requires domain knowledge. Furthermore, the learned features from each dataset are relatively independent and lack of transferability. The correlation between these extracted features is omitted.

Therefore, in light of the practical implementation of NILM, it is desirable to transfer the knowledge learned from one area (the source domain) with labeled data to another area (the target domain). We expect the learned knowledge can be transferred within different buildings in the same dataset or across the datasets. Different from most of the traditional machine learning methods, the transfer learning methods do not require that the training and testing samples are independent and identically distributed. On this basis, it focuses on how to transfer the gained knowledge from solving one problem and apply it to a related problem [14]. Reference [15] attempts to factor out the heterogeneity in homes and weather across regions to build a region independent model, under the constraints that the target domain should have a similar set of appliances as the source domain. In [13], [16], [17], deep learning based models with transferability are proposed for NILM. They assume that those features exacted by neural networks are invariant across appliances and data domains. However, because of the representation bias in the distribution of the source and target domains [18], directly fine-tuning the deep network's parameters on a small number of target labeled data is problematic [19]. The differences in the distribution of representations should be considered and measured. It is necessary to optimize the domain invariance by finding a representation that makes domains become similar, and simultaneously reduce the domain discrepancy to enhance the feature transferability.

To address the above challenges, this paper proposes a new CNN architecture with knowledge transferability by optimizing the regression loss on the labeled energy consumption data as well as domain adaptation loss on the source

domain and target domain. The main contributions of this work are threefold: 1) A novel end-to-end transfer learning network (TLN) for NILM is proposed, which can take advantage of both the labeled data from the source domain and the unlabeled data from the target domain. 2) A sequence-to-sequence modeling architecture constructed by stacking temporal convolutional blocks and residual connections is proposed to better learn the dynamic load signatures of individual appliances. 3) A domain adaptation loss is proposed for quantifying the domain distribution discrepancy between the source and target domain examples. By jointly optimizing domain adaptation and energy disaggregation loss, a more accurate invariant representation for the evolution of appliances states can be learned, and thus the performance and transferability of the proposed method are improved. Data experiments on ground truth data validate both the accuracy and the robustness of the proposed model.

The rest of this paper is organized as follows. Section II describes the deep domain adaptation and presents the problem formulation. Section III introduces the framework of TLN for NILM. Section IV presents the experimental setup. The experimental results and discussions are shown in Section V, and conclusions are drawn in Section VI.

II. PROBLEM STATEMENT

A. Deep Domain Adaptation

The collected customers' energy consumption data from different domains are under different distributions. This discrepancy in the distribution of the source and target data representations becomes the major obstacle in applying NILM models across domains. By exploring domain-invariant features that bridge different domains of substantial distribution discrepancy, domain adaptation process can help transfer the knowledge learned from the labeled source domain to the unlabeled target domain [20]. The domain invariant features are learned in [21] by measuring the maximum mean discrepancy (MMD). Although the distribution mismatch between domains in the latent space is reduced, the performance is limited by its shallow two-layer network. Reference [19] proposes a deep domain confusion network and improves the work of [21] by adding an adaptation layer and an additional domain confusion loss. However, a single adaptation layer restricts the transferability and scalability. The number and position of the adaptation layer should not be determined empirically.

Therefore, an improved deep domain adaptation transfer learning network for NILM is proposed. The hidden representations of all the task-specific layers are included to calculate the domain discrepancy. A novel domain adaptation loss is introduced. The transferability and generalization ability are improved by jointly reducing the domain adaptation loss and load disaggregation loss.

B. Domain Adaptation for NILM

The goal of NILM problem is to extract the energy consumption of individual appliances from the whole-house consumption data. For a household with N appliances, let $\mathbf{x} =$

$(x_1, x_2, \dots, x_T)$ and $\mathbf{y}^i = (y_1^i, y_2^i, \dots, y_T^i)$ denote the whole-house load and the usage of the i -th appliance, respectively. the whole-house load at time t can be expressed as:

$$x_t = \sum_{i=1}^N y_t^i + \varepsilon_t \quad (1)$$

where ε_t represents the measurement noises. When the j -th appliance is the target appliance, for the traditional supervised learning methods, a supervised regression model $\mathcal{R}(\mathbf{x}, \mathbf{y}^j)$ can be trained by using the labeled household data. However, the performance of this trained model is generally not ideal on an unseen household which is not present in the training dataset [13]. This lower generalization ability is due to the distribution discrepancy between the training data and testing data. To this end, we use the domain adaptation method to make full use of the unlabeled data, and thus the model transferability to different households can be improved.

Suppose the source domain dataset $\mathcal{D}_S = \{(x_S^1, y_S^1), (x_S^2, y_S^2), \dots, (x_S^{n_s}, y_S^{n_s})\}$ includes the labeled data, and the target domain dataset $\mathcal{D}_T = \{x_T^1, x_T^2, \dots, x_T^{n_t}\}$ includes the unlabeled data, where $x_S^i \in \mathbf{X}_S$ is the input whole-house load of the source domain, $y_S^i \in \mathbf{Y}_S$ is the usage of the appliance as the label, and $x_T^i \in \mathbf{X}_T$ is the input whole-house load of the target domain without labels. The number of source and target data are n_s and n_t , respectively. The feature space and the label space of the source domain are respectively denoted as $\mathbf{X}_S$ and $\mathbf{Y}_S$, and the feature space of the target domain is denoted as $\mathbf{X}_T$. In our paper, the domain distribution is defined as the marginal probability of the feature space [14]. In practice, the marginal probability distribution of target domain is different from the source domain, that is $P_S(\mathbf{X}_S) \neq P_T(\mathbf{X}_T)$ [14], [16]. These distribution differences might reduce the generalization ability of the model $\mathcal{R}(\mathbf{X}_S, \mathbf{Y}_S)$, since it ignores the distribution characteristics of the target domain. To mitigate the influence of domain discrepancy between domains, this paper aims to train a load disaggregation regressor $\mathcal{R}(\mathbf{X}_S, \mathbf{Y}_S, \mathbf{X}_T)$, considering both the labeled source domain dataset and unlabeled target domain dataset. The feature transferability can be enhanced by reducing the domain discrepancy.

III. MODEL FRAMEWORK

In this section, the proposed methodology for disaggregating the appliance usage from the whole-house energy consumption data is described. Different from most of the standard deep learning methods which are examined in the same domain, the proposed TLN is designed to learn a deep representation that is both domain invariant and which offers strong separation ability. The network architecture of TLN is shown in Fig. 1. Given the observed whole-house load data from labeled source domain and from unlabeled target domain, the high-level features are extracted by the temporal convolutional layers. Then the supervised regression loss can be calculated using the labeled source signals, while the domain adaptation loss can be obtained by comparing the distance between the source and target domain distributions. These losses are combined to determine the weights of the neural network using

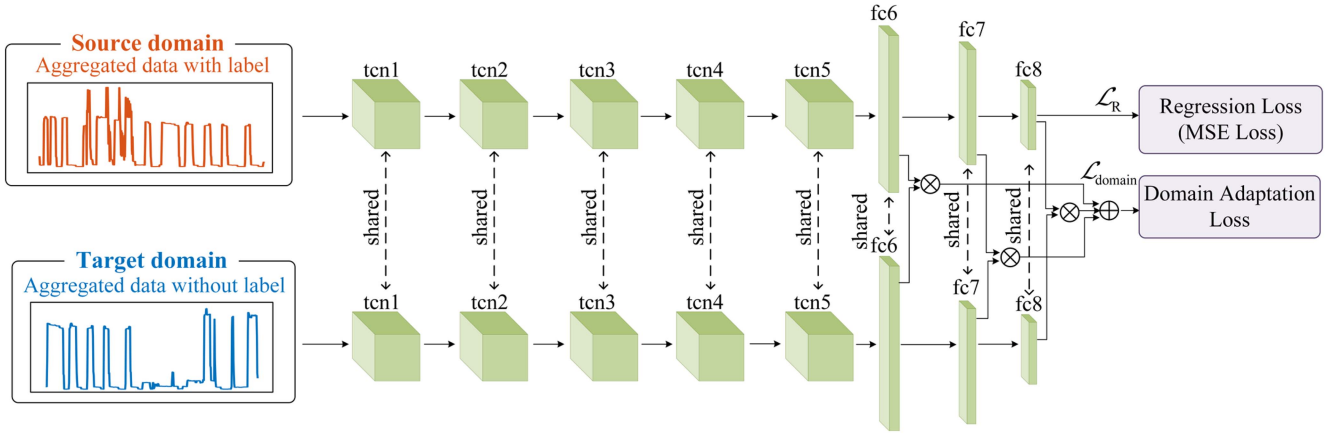


Fig. 1. The architecture of the proposed TLN for NILM.

backpropagation for training. Note that the weights of parameters are shared between the upper source part and the lower target part. The network is jointly trained to minimize the total loss. Each part of the proposed framework is detailedly described in the following subsections.

A. Temporal Convolutional Network for Feature Extraction

The temporal convolutional network (TCN) has already shown its success for model training and achieved better performance compared with recurrent networks across a various range of modeling tasks [22]. On the one hand, the causal convolutions in the model differ from the standard convolutions by using only the samples from the previous time steps to estimate the current output, and thus guarantee that there is no information leakage from the future into the past. Maintaining this causality is important for NILM problem since it can estimate the usage of appliances in real-time. With these real-time disaggregation results, the retailer can suggest the customers deferring the usage of energy-intensive appliances from peak periods to valley periods, which can both save their electricity bills and reduce the load on the network. On the other hand, the TCN can take a sequence of any length and produce the output with the same length as the input [12].

In order to accomplish the first point, causal convolutions are implemented in the TCN, where the output at time t is only convolved with elements at time t and earlier in the previous layer. To achieve the second point, a 1-D fully convolutional network is used where the input layer and each hidden layer have the same length, and zero padding of $\text{kernel size} - 1$ length is added to keep the subsequent layers have the same length with the previous ones. This basic architecture can only look back the history information, whose size is linear with the depth of the network. Thus, the dilated causal convolutions are introduced following the work of [23] and [24], which can support exponential expansion of the receptive field without loss of resolution or coverage.

A dilated causal convolution is a convolution process that applies the filter over an area larger than its length by skipping input values with a certain step. More specifically, given a 1-D input data $x \in \mathbb{R}^n$ and a filter $f : \{0, 1, \dots, k-1\} \rightarrow \mathbb{R}$, the

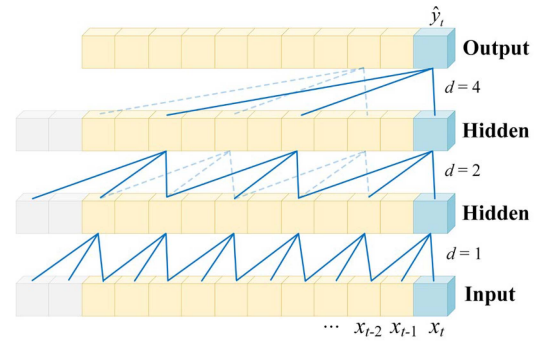


Fig. 2. Visualization of dilated casual convolution layers.

dilated causal convolution operation F on element s of the sequence can be defined as follows:

$$F(s) = (x *_d f)(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-d \cdot i} \quad (2)$$

where d is the dilation factor and $*_d$ represents the d -dilated convolution; k denotes the filter size, and $s - d \cdot i$ indicates the direction of the past, different from the $s - i$ for the standard convolution. Therefore, the dilation is similar to setting fixed step between each pair of adjacent layers. Fig. 2 depicts a dilated convolution with dilation factors $d = 1, 2, 4$ and filter size $k = 3$. When $d = 1$, a dilated causal convolution becomes a standard convolution. The obtained receptive field of one layer is $(k-1)d$, thus it can be increased by choosing larger filter size k and dilation factor d . Usually, we increase d exponentially for each layer i of the network, that $d = 2^{i-1}$. This stacking dilated causal convolution architecture enables the deeper level outputs representing extremely large history inputs with just a few layers, which is effective and saves computational cost.

Another way of enriching the TCN receptive field is to construct deeper and wider layers of neural networks. However, the degradation problem often occurs when deeper stacking networks begin converging. The accuracy gets saturated at first, and then degrades rapidly as the network depth increases. Facing that problem, a deep residual

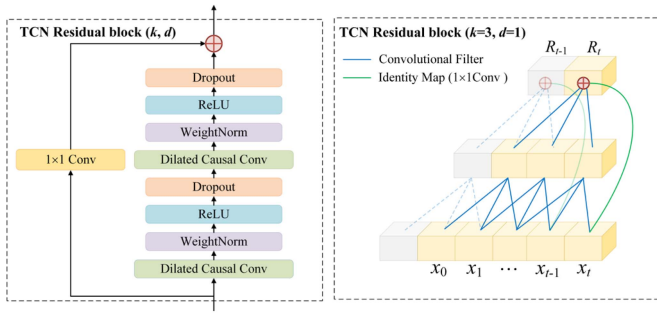


Fig. 3. The TCN architecture. (a) a TCN residual block, (b) an example of the TCN residual block with filter size $k = 3$, and dilation factor $d = 1$. The blue lines represent the filters in the residual function and the green line is identity mapping (1×1 conv).

learning framework (ResNet) [25] is proposed using residual representations and shortcut connections. On the basis of ResNet, TCN stacks the dilated causal convolution layers by a residual mapping. An example of the TCN residual block is shown in Fig. 3(a), including two dilated causal convolution layers with ReLU activation. Weight normalization for speeding up the network convergence and dropout for preventing network overfitting are included in the block. Fig. 3(b) illustrates an TCN residual block. To ensure the output is in the same length with the input, an additional 1×1 convolution layer is added in TCN that the element-wise $\oplus$ addition receives the same shape tensors, which is different from the standard ResNet that the input is added directly to the output. In the TCN residual block, the outputs of a series of transformations $\mathcal{F}$ are added to identity mapping of input x , which is represented as:

$$R(s) = \mathcal{F}(x, \theta) + x \quad (3)$$

where θ denotes the set of parameters in the network. This way of connection can learn the modifications to the identity mapping rather than the entire transformation, which has already been demonstrated its effectiveness. Suppose the input of the p -th block in a shallower location is x_p , then the output x_q of q -th block in a deeper location through forward propagation can be calculated as:

$$x_q = x_p + \sum_{i=p}^q \mathcal{F}(x_i, \theta_i) \quad (4)$$

where θ_i is the set of parameters of i -th block. Denoting the loss function as $\mathcal{L}$, the backpropagation using chain rules can be formulated as

$$\frac{\partial \mathcal{L}}{\partial x_p} = \frac{\partial \mathcal{L}}{\partial x_q} \left(1 + \frac{\partial}{\partial x_p} \sum_{i=p}^{q-1} \mathcal{F}(x_i, \theta_i) \right) \quad (5)$$

The gradient $(\partial \mathcal{L} / \partial x_p)$ consists two parts. One is $\frac{\partial \mathcal{L}}{\partial x_q}$ that propagates information directly back to any shallower block p without concerning any weight layers [26]. The other one is $\frac{\partial}{\partial x_p} \sum_{i=p}^{q-1}$ that propagates through the weight layers. In general, this term cannot always be equal to -1 for all samples in a mini-batch, which ensures the gradients are less likely to vanish when the weights are small.

B. Domain Adaptation Loss

For cross-domain transfer learning, it is essential to reduce the data distribution discrepancy between the source and target domain in order to learn the domain invariance representation. Several approaches have been proposed to measure the data distribution distances between two domains. Among them, two commonly used domain adaptation methods are the MMD [21] and the deep CORAL loss [27], which are simple yet effective, and can be easily implemented in practice. Therefore, we take these two methods to search for a representation that minimizes the distribution distance between domains. Given a source domain dataset $\mathcal{D}_S$ and a target domain dataset $\mathcal{D}_T$. Then an empirical approximation formula of the MMD distance is computed as follows:

$$MMD(X_S, X_T) = \left\| \frac{1}{n_s} \sum_{i=1}^{n_s} \phi(x_S^i) - \frac{1}{n_t} \sum_{j=1}^{n_t} \phi(x_T^j) \right\|_{\mathcal{H}} \quad (6)$$

where $\phi(\cdot)$ is a feature space mapping function that turns the original feature space into the reproducing kernel Hilbert space $\mathcal{H}$. Here, we choose the Gaussian kernel as the kernel function.

The deep CORAL method is another popular criterion that aligns the second-order statistic correlation information of the source and target distributions with a non-linear transformation, in order to minimize the difference in learned feature covariance across domains.

$$\mathcal{L}_{CORAL} = \frac{1}{4L^2} \|C_S - C_T\|_F^2 \quad (7)$$

where $\|\cdot\|_F^2$ represents the squared matrix Frobenius norm; L is the dimension of the features; C_S and C_T are the covariance matrices of the source and target samples denoted by:

$$C_S = \frac{1}{n_s - 1} \left(X_S^T X_S - \frac{1}{n_s} (\mathbf{1}^T X_S)^T (\mathbf{1}^T X_S) \right) \quad (8)$$

$$C_T = \frac{1}{n_t - 1} \left(X_T^T X_T - \frac{1}{n_t} (\mathbf{1}^T X_T)^T (\mathbf{1}^T X_T) \right) \quad (9)$$

where $\mathbf{1}$ is a column vector with all elements equal to 1. Suppose X_S^{ij} and X_T^{ij} indicate the j -th dimension of the i -th source and target data, respectively. The gradient related to the input features can be calculated using the chain rule as follows:

$$\frac{\partial \mathcal{L}_{CORAL}}{\partial X_S^{ij}} = \frac{1}{L^2(n_s - 1)} \times \left(\left(X_S^T - \frac{1}{n_s} (\mathbf{1}^T X_S)^T \mathbf{1}^T \right)^T (X_S - X_T) \right)^{ij} \quad (10)$$

$$\frac{\partial \mathcal{L}_{CORAL}}{\partial X_T^{ij}} = \frac{1}{L^2(n_t - 1)} \times \left(\left(X_T^T - \frac{1}{n_t} (\mathbf{1}^T X_T)^T \mathbf{1}^T \right)^T (X_S - X_T) \right)^{ij} \quad (11)$$

In order to fully take advantage of the superior ability of both domain adaptation methods, we combine the MMD and the CORAL loss to make it more adaptive to high dimensional

data divergence calculation.

$$\mathcal{L}_{\text{domain}}(Z_S, Z_T) = \sum_{l=l_1}^{l_2} \mu \text{MMD}^2(Z_S^l, Z_T^l) + (1 - \mu) \mathcal{L}_{\text{CORAL}}(Z_S^l, Z_T^l) \quad (12)$$

where $\mu \in [0, 1]$ is a weighting factor to regularize these two distribution distances; Z_S^l and Z_T^l are the feature representation of l -th layer for the source and target domain, respectively. In our implementation, we set $l_1 = 6$ and $l_2 = 8$.

C. Model Training for Knowledge Transfer

As shown in Fig. 1, our aim is to learn a representation by minimizing the distance between the source and target data distribution, and train an estimator on the source domain with labeled data, which can then be directly applied to the target domain with minimal loss in accuracy. In this paper, we extend the classic CNN architecture called Alexnet [28], by stacking the TCN blocks instead of the standard convolution layers. It is comprised of five TCN residual blocks (*tcn1-tcn5*) and three fully connected layers (*fc6-fc8*) in the proposed TLN model. Besides, the ReLU activation and the dropout technique are also used in *fc6-fc8* to improve the model performance. As demonstrated in [29], the convolution layers can learn general representation features which could be transferable in the first layer to the third layer, and there exists slight domain bias in the fourth and the fifth layer, but the performance significantly drops on *fc6-fc8*. The *fc* layers are only aimed at their original target task, which are task-specific and cannot be directly fine-tuning on the target domain with limited labeled data. Thus, when adapting the classic Alexnet to our TLN, we share the first five TCN blocks *tcn1-tcn5* to preserve the efficacy of feature transferability. Moreover, instead of adapting a single layer, we take multi-layer adaption *fc6-fc8*, which can bridge both the marginal distribution and the conditional distribution for domain discrepancy.

To evaluate the performance of load disaggregation, the ground truth data y and prediction results $\hat{y}$ of source domain are used for the supervised regression loss calculation, which is defined as:

$$\mathcal{L}_R(y, \hat{y}) = \frac{1}{n_s} \sum_{i=1}^{n_s} J(\theta(x_i), y_i) = \frac{1}{n_s} \sum_{i=1}^{n_s} (y_i - \hat{y}_i)^2 \quad (13)$$

where $J(\cdot)$ is the mean square error (MSE) loss function; and $\theta(x_i)$ is the conditional probability outputs of x_i .

After training TCN on source labeled data and extracting the suitable features, it is required to measure the hidden representations of *fc6-fc8* to adjust the source and target distributions to become similar. This can be realized by adding the domain adaptation loss to the objective function of TLN as follows:

$$\min_{\Theta} \mathcal{L} = (1 - \lambda) \mathcal{L}_R(y, \hat{y}) + \lambda \mathcal{L}_{\text{domain}}(Z_S, Z_T) \quad (14)$$

where Θ denotes the set of all TLN parameters; and $\lambda \in [0, 1]$ is a weighting parameter using for regularizing the scale of the losses. It should be noted that a high λ means the domain adaption loss is dominant in the final loss function with the

expense of degraded regression accuracy, while a small λ can lower the impact of feature transferability.

On the basis of the domain adaptation, we can fully exploit both labeled data from the source domain and unlabeled data from the target domain by minimizing the total loss function of TLN. The parameters Θ for the network are achieved by backpropagation. An efficient mini-batch gradient-based stochastic optimization algorithm called Adam optimizer is employed for training process. The early-stopping criterion is also introduced to avoid overfitting. The source domain data is randomly selected, and the limited target data is reused to increase the generalization ability of the network. The number of every batch of source and target domain data are the same.

IV. EXPERIMENTAL SETUP

A. Dataset Preparation

1) Dataset: Three publicly available datasets, REDD (R) [30], UK-DALE (U) [31] and REFIT (F) [32], are used to evaluate the performance of the proposed methods for NILM.

The REDD dataset consists of appliance-level energy consumption data sampled every 3 seconds and whole-house measurements sampled every 1 second. This dataset is collected from 6 real buildings in USA [30]. The length of measurements is between 3 and 19 days for different buildings.

The UK-DALE dataset collects smart meter data of 5 buildings in U.K., which contains more load profiles compared with the REDD. The sampling resolution of the whole-house consumption and the corresponding appliance-level consumption are 1 second and 6 seconds, respectively. The period of this dataset covers 39 to 600 days [31].

The REFIT dataset is the largest dataset among these three datasets. It contains load profiles from 20 households in England with the time spanning from 2013 to 2015. The active power of both whole-house and individual appliance consumption are recorded at a resolution of 8 seconds [32].

To keep consistency with the resolution of the whole-house and appliance-level measurements for these three datasets, we set the resolution to 6 seconds in this analysis. Therefore, we up-sample the data of REDD and the whole-house data of UK-DALE, and down-sample the data of REFIT to 6-second level.

For a fair comparison with the previous works [13], [16], the energy disaggregation results of five representative appliances are reported in this paper, which are the washing machine (WM), the dishwasher (DW), the microwave (MV), the fridge (FG) and the kettle (KT). Note that the kettle is not recorded in the REDD dataset. These selected appliances cover different power patterns and power levels. The load patterns of the KT are simple, whereas the profiles of WM are complicated. The FG works at small power consumption, whereas load profiles of the others are relatively large. Moreover, The power consumption of these 5 studied appliances accounts for a large proportion of the power consumption in a building. The appliances with particularly low power, such as lighting devices, will not be selected as the target appliances, since the effect of these appliances on the whole-house load tends to get lost in the noise. Thus, these 5 studied appliances are generally representative.

2) Data preprocessing: Data cleansing process is applied at first to omit the subsequences of load profiles with missing or misread readings over 5 minutes. To reduce the influence of the sample imbalance on the model performance, we randomly select on-state samples and off-state samples from training datasets and keep the number of on-state samples accounts for more than 40%. Then a sliding window T with stride length l_s runs forward the sequences to obtain the input sample $x = (x_1, x_2, \dots, x_T)$. The principle of choosing the window length is that it must exceed the length of one duty circle of each appliance. Then, the model can learn the full dynamic load patterns from the input windows. In this paper, the proposed TLN network is a sequence-to-sequence architecture. With the input $x = (x_1, x_2, \dots, x_T)$, the output sequence of the i -th appliance is $y^i = (y_1^i, y_2^i, \dots, y_T^i)$. There are two reasons for choosing the sequence-to-sequence architecture instead of the sequence-to-point architecture: 1) Due to the structure of dilated causal convolution as shown in Fig. 2, only the last point of window could be used as the point output. However, as mentioned in [6], the performance of the windows where the element is near the edges is worse than that near the midpoint. 2) We do not use any future information to estimate the electricity consumption of appliances at time T . This sequence-to-sequence architecture prevents information leakage from the future, and makes the proposed method more suitable for real-time practical applications.

The building 3 from REDD, the building 2 from UK-DALE and the building 2 from REFIT are used as the testing dataset and the target domain. Only the whole-house consumption of each building is available, whereas the actual electricity consumption of individual appliances is unknown.

B. Evaluation Metrics

Two widely used metrics are adopted to quantitatively evaluate the performance of the proposed model.

- 1) Mean absolute error (MAE):

$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T |y_t^i - \hat{y}_t^i| \quad (15)$$

where y_t^i and $\hat{y}_t^i$ are the ground truth and the predicted value of the i -th appliance at time t , respectively.

- 2) Signal aggregate error (SAE):

$$\text{SAE} = \frac{1}{T_L} \left| \sum_{t=1}^{T_L} y_t^i - \sum_{t=1}^{T_L} \hat{y}_t^i \right| \quad (16)$$

where T_L is the number of time points within a period. Different from MAE that measures the mean of error at each time point, SAE measures the total estimation error over a longer period of time. We set T_L to 3600, which corresponds to the number of data points in 6 hours with 6s sampling frequency.

C. Experimental Setup

1) Implementation configuration: We set the sliding window length T to 600. The stride length l_s is set to 8 for the UK-DALE dataset and the REFIT dataset. Due to the limited

TABLE I
OPTIMAL TRAINING PARAMETERS FOR THE MODEL

Parameter description	Optimal value
Number of TCN blocks	5
Number of fully connected layers	3
Number of filters in each TCN block	128
Filter size	3
Dilation factor	2^i for block i
Activation function	ReLU
Weighting factor μ	0.4
Weighting factor λ	0.6
Number of maximum epochs	30
Mini-batch size	256
Number of minimum early-stopping epochs	4
Learning rate	0.01

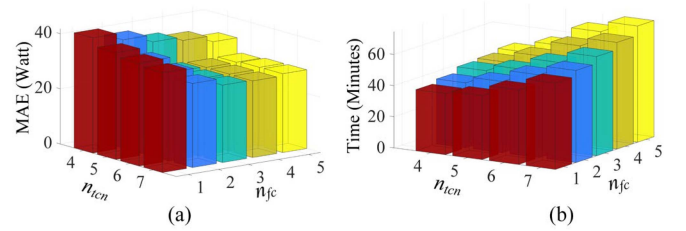


Fig. 4. Comparison of MAE (left) and training time (right) across different number of shared TCN layers n_{tcn} and fully connected layers n_{fc} .

data in the REDD dataset, we set l_s to 2 to obtain enough samples for model training. The parameters of the proposed TLN model are listed in Table I.

2) Comparison methods: To compare the performance of the proposed TLN model with other models, the Alexnet is regarded as the baseline method by removing all the domain adaptation components of the proposed TLN model. In addition, three state-of-the-art NILM methods with transferability are used for comparison. For simplicity, we call the CNN method in [13], gated recurrent unit (GRU) network in [13] and the sequence-to-point method in [16] as T-CNN, T-GRU, TS2P, respectively. All these algorithms are implemented in Python with the packages of Pytorch, and are run on a standard PC with an NVIDIA GTX1650 GPU.

V. CASE STUDIES AND DISCUSSIONS

A. Different Settings Related to the Hyperparameters

Selecting appropriate hyperparameters is essential to improve the model performance. Since it is more challenging to generalize a deep learning model trained on a limited dataset to other unseen buildings, we use the REDD dataset, which is the smallest among the three datasets, to tune the key parameters of the proposed model. We evaluate the model performance on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$, where the superscript number denotes the index of the building in the dataset, and the superscript A denotes all the buildings except the test building.

Selection of different number of layers: Two important hyperparameters that influence the disaggregation performance of the proposed method are the number of shared TCN layers n_{tcn} and the number of fully connected layers n_{fc} . Here, we set $n_{tcn} \in \{4, 5, 6, 7\}$ and $n_{fc} \in \{1, 2, 3, 4, 5\}$. Fig. 4 shows

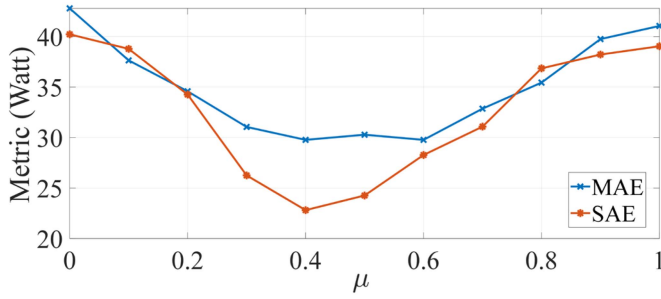


Fig. 5. Comparison of different weighting factor μ on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$.

the average MAE and the average training time of four appliances on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$ using the proposed TLN model. As shown in Fig. 4(a), when n_{tcn} and n_{fc} increase, the MAE descends at first and then flattens out gradually. Although increasing n_{tcn} and n_{fc} can improve the model performance, it will introduce more parameters to be trained and thus increase the runtime of the training process, as shown in Fig. 4(b). In addition, too many TCN layers and fully connected layers may also cause overfitting problems. For example, when $n_{tcn} = 7$ and $n_{fc} = 5$, the model performance decreases with the rise of MAE. Therefore, there is a trade-off between the disaggregation accuracy and the total training time. In this paper, we set n_{tcn} and n_{fc} as 5 and 3, respectively. Practitioners can use these two figures to determine a suitable combination of n_{tcn} and n_{fc} .

Selection of weighting factor μ : The tuning process of the weighting factor μ is illustrated in (12). As discussed in Section III, μ is used to adjust the scale of two different domain distances. The proposed model is tested by using different values of μ for each appliance on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$. The average results of MAE and SAE for four appliances are shown in Fig. 5. As shown, the performance of taking both domain distances into consideration is much better than that of using only one type of domain distance (see both edges of $\mu = 0$ and $\mu = 1$ in Fig. 5). This observation can well demonstrate the effectiveness of taking these two distances to search for feature representations to minimize the distribution distance between domains. It can also be observed that the model achieves the best results at the point of $\mu = 0.4$. For this reason, the weighting factor μ is chosen as 0.4 in this paper.

Selection of weighting factor λ : Another important parameter is the weighting factor λ in (14), which determines the proportion of the domain adaptation loss in the total loss. The model performance is evaluated for the washing machine on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$. Fig. 6 shows the MAE, the total loss and the domain adaptation loss. The improvement in disaggregation accuracy can be observed when the weighting factor λ is adjusted between the regression loss and the domain adaptation loss. If λ is set too low, the domain adaptation will have no effect on the learned representation, but if λ is set too high, the representation will be too similar to a heavy regularization. Therefore, a suitable trade-off between them can enhance the feature transferability and improve the energy disaggregation accuracy. Since the best performance can be achieved when $\lambda = 0.6$, we set λ to 0.6 for all experiments in this

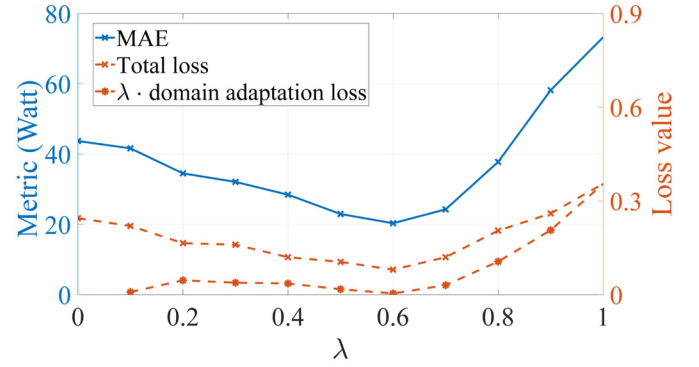


Fig. 6. Comparison of different weighting factor λ on the task of $\mathbf{R}^A \rightarrow \mathbf{R}^3$.

paper. It is also observed that the individual performance using one loss function is not satisfactory as shown at the edges of Fig. 6. As mentioned in Section III, the learned features need to be not only discriminative to train a strong classifier, but also invariant between the source domain and the target domain. If only the regression loss is minimized, it is likely to cause overfitting issues to the source domain and reduce the performance on the target domain. On the other hand, minimizing the domain adaptation loss alone will result in the degeneration of extracted features. Therefore, jointly optimizing the regression loss and domain adaptation loss is more likely to learn the hidden features that work well on the target domain.

B. Domain Adaptation Within the Same Dataset

In this subsection, one building is used as the target domain, and the other buildings in the same dataset as the source domain. For each testing appliance, we randomly select 100 sample windows for model testing. Table II shows the average and standard deviation of the performance indices across different testing sample windows for the five disaggregation methods. The row Improvement shows the relative improvements in the average indices of the proposed TLN method compared with the best result of the other four methods. It is shown that the proposed method can achieve lower MAE and SAE than other methods for most of the appliances. We further apply a t-test with 95% confidence level on the evaluation metrics of each pair of the proposed model and another benchmark model. The t-test results show significant difference between the average values. These statistical results demonstrate a superior disaggregation ability of the proposed method. Although the TS2P performs better in some cases, it still needs some labeled data of appliances from the target domain to fine-tune its model parameters. Without an appropriate fine-tuning process, the generalization ability of this model may not be guaranteed. Different from the TS2P model, the electricity consumption of individual appliances in the target domain is completely invisible to the proposed model during the model training stage. Therefore, the proposed method is more scalable to practical applications in that the trained model can be directly employed to an unseen building with a high disaggregation accuracy.

TABLE II
COMPARISON OF PERFORMANCE INDICES ON THE DOMAIN ADAPTATION WITHIN THE SAME DATASET

Appliance	Method	REDD ($R^A \rightarrow R^3$)		UK-DALE ($U^A \rightarrow U^2$)		REFIT ($F^A \rightarrow F^2$)	
		MAE (Watt)	SAE (Watt)	MAE (Watt)	SAE (Watt)	MAE (Watt)	SAE (Watt)
WM	Baseline	25.11±9.32	28.32±8.47	20.03±10.05	23.21±7.71	22.34±13.02	25.32±14.29
	T-CNN	23.85±7.20	25.84±5.98	17.96±8.82	19.18±9.92	19.78±9.04	21.07±12.32
	T-GRU	23.42±7.63	26.16±5.44	17.75±7.78	18.62±8.57	20.11±10.11	22.43±10.43
	TS2P	21.96±5.33	24.33±4.86	16.92±6.65	18.04±6.44	18.34±7.26	19.42±8.60
	Proposed	19.89±3.09	23.08±3.35	15.85±4.43	17.21±5.12	17.95±5.58	18.71±6.44
	Improvement	9.4%	5.1%	6.3%	4.6%	2.1%	3.7%
DW	Baseline	28.27±13.33	27.79±10.51	34.44±15.25	32.51±14.14	17.95±8.57	16.26±7.25
	T-CNN	25.82±14.48	26.86±11.24	32.87±12.25	31.17±11.01	16.89±6.25	15.97±8.05
	T-GRU	26.37±15.62	28.11±11.85	33.52±14.44	32.95±13.66	17.07±6.37	16.26±7.62
	TS2P	23.04±9.77	19.85±8.86	30.08±11.25	28.58±10.87	15.11±6.85	15.32±5.50
	Proposed	23.75±9.91	20.74±7.74	28.4±9.95	26.66±8.84	15.35±5.77	14.85±5.23
	Improvement	-3.1%	-4.5%	5.6%	6.7%	-1.6%	3.1%
MV	Baseline	35.75±11.58	35.01±9.52	13.51±6.71	12.85±7.15	17.35±6.34	16.88±7.08
	T-CNN	35.11±14.41	36.27±10.55	10.82±4.32	11.18±5.30	16.11±7.11	16.58±8.32
	T-GRU	34.25±13.58	36.95±11.03	11.34±5.86	11.65±4.27	16.25±7.95	16.32±6.19
	TS2P	31.23±9.74	30.09±6.32	9.65±5.06	9.21±5.63	13.72±6.41	12.23±7.75
	Proposed	29.94±6.12	29.61±4.99	9.74±4.84	8.98±6.25	12.19±5.32	11.77±6.61
	Improvement	4.3%	1.6%	-0.9%	2.6%	12.6%	3.9%
FG	Baseline	40.05±11.78	42.52±10.14	27.58±8.33	24.19±7.74	29.04±8.92	27.87±6.05
	T-CNN	37.41±13.06	36.32±11.18	26.19±7.04	25.32±7.35	27.33±6.04	26.54±5.26
	T-GRU	38.72±12.11	37.59±12.35	25.77±8.15	26.77±6.85	27.75±7.17	27.01±7.54
	TS2P	32.74±9.25	34.08±8.66	22.88±5.30	21.96±4.33	24.39±5.35	22.21±5.41
	Proposed	31.29±6.55	33.04±7.15	22.66±5.11	21.44±4.68	23.2±5.61	22.38±4.25
	Improvement	4.4%	3.1%	2.7%	2.4%	4.9%	-0.8%
KT	Baseline	-	-	12.77±5.81	11.05±4.92	10.97±6.37	9.85±5.24
	T-CNN	-	-	10.85±4.41	10.31±4.13	9.16±5.22	8.95±5.31
	T-GRU	-	-	11.02±4.77	10.75±4.26	9.45±5.44	9.42±5.09
	TS2P	-	-	8.53±3.31	7.92±3.81	7.38±4.98	6.81±3.41
	Proposed	-	-	7.68±3.38	6.88±3.25	6.95±4.56	6.76±3.27
	Improvement	-	-	10.0%	13.1%	5.8%	0.7%

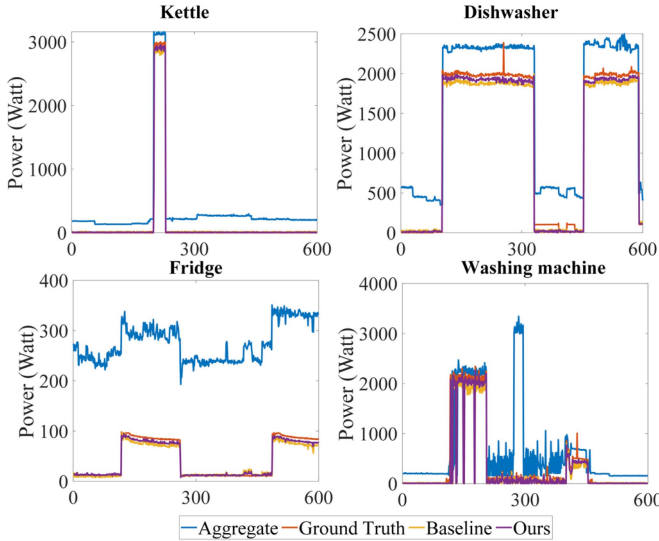


Fig. 7. Examples of the disaggregation results of kettle (top left), the dishwasher (top right), the fridge (bottom left) and the washing machine (bottom right) on the task of $U^A \rightarrow U^2$.

Fig. 7 plots examples of the disaggregation results of the kettle, the dishwasher, the fridge and the washing machine on the task of $U^A \rightarrow U^2$ with the ground truth. As can be seen, the proposed model can better separate the appliances' electricity consumption from the whole-house load data than the baseline method, especially for the washing machine with complicated state patterns. It can also be observed that when the target appliance is not switched on, its run

TABLE III
COMPARISON OF EVALUATION METRICS WHEN TARGET APPLIANCE IS NOT SWITCHED ON IN THE TARGET DOMAIN ($U^A \rightarrow U^2$)

Appliance	Method	MAE (WATT)	SAE (WATT)
WM	Baseline	3.11±1.13	3.45±1.88
	proposed	0.58±0.23	0.66±0.19
DW	Baseline	5.75±2.35	6.02±2.56
	proposed	1.14±0.21	1.38±0.38
MV	Baseline	2.77±0.95	2.49±1.10
	proposed	0.85±0.33	0.75±0.23
FG	Baseline	8.22±2.77	7.67±3.65
	proposed	2.78±1.40	2.98±1.44
KT	Baseline	1.89±0.51	2.24±0.87
	proposed	0.78±0.41	1.05±0.46

is rarely over-disaggregated which might be contributed by other loads. Furthermore, it is interesting to investigate the performance of the proposed model when the target appliance is not switched on in the target domain. We conduct this experiment on the task of $U^A \rightarrow U^2$. The measurements of each target appliance of U^2 are respectively set to zero, and the whole-house load should minus the original readings of the target appliance. The data of U^A remains the same. We randomly select 100 window samples for each target appliance from the test dataset. Table III presents the average and standard deviation results across selected window samples. As shown, the runs of target appliance are not over-disaggregated using the proposed model. The reason for the superior performance is that different appliances have different specific load patterns and power levels.

TABLE IV
COMPARISON OF PERFORMANCE INDICES ON THE DOMAIN ADAPTATION ACROSS DATASETS

Appliance	Method	$F^A \rightarrow U^2$		$U^A \rightarrow R^3$		$R^3 \rightarrow F^2$	
		MAE (Watt)	SAE (Watt)	MAE (Watt)	SAE (Watt)	MAE (Watt)	SAE (Watt)
WM	Baseline	27.66±11.12	25.11±6.45	38.12±13.11	36.26±10.07	50.19±16.55	47.07±15.57
	T-CNN	24.85±8.18	23.68±6.85	36.06±11.25	37.17±8.75	47.25±12.72	48.22±13.02
	T-GRU	23.97±7.74	24.7±5.78	34.41±12.02	36.45±7.77	46.83±13.31	44.39±14.11
	TS2P	20.04±5.22	21.04±5.11	30.04±8.57	28.18±6.15	44.12±10.88	40.28±10.81
	Proposed	18.87±5.86	18.71±5.80	29.87±6.25	26.66±6.87	33.58±10.58	30.04±11.09
	Improvement	5.8%	11.1%	0.6%	5.4%	23.9%	25.4%
DW	Baseline	42.55±7.58	39.23±8.96	44.98±12.42	42.02±13.47	53.17±17.17	51.42±16.36
	T-CNN	38.74±8.65	36.65±7.74	41.45±9.22	39.74±10.01	47.35±13.52	44.33±14.24
	T-GRU	36.69±8.93	38.16±8.18	43.35±10.07	42.82±9.94	49.09±15.10	48.52±15.32
	TS2P	35.08±6.85	34.5±6.99	38.77±8.41	36.11±6.87	44.52±11.85	38.8±12.11
	Proposed	30.18±5.85	28.49±6.24	35.04±7.75	31.76±7.33	34.75±9.58	30.42±13.04
	Improvement	14.0%	17.4%	9.6%	12.0%	21.9%	21.6%
MV	Baseline	20.58±9.58	18.08±8.74	50.14±13.30	51.32±14.35	56.85±19.22	52.77±20.02
	T-CNN	18.74±9.05	17.49±8.31	48.87±10.05	46.32±12.21	49.05±18.71	45.57±16.64
	T-GRU	16.99±8.74	17.68±9.24	51.95±11.71	48.08±10.58	53.22±17.09	50.19±15.85
	TS2P	13.21±6.74	15.72±6.60	43.28±9.25	39.95±9.47	40.17±13.30	38.12±10.11
	Proposed	13.02±7.21	14.04±7.06	39.69±7.75	37.17±10.62	36.64±12.17	31.17±10.27
	Improvement	1.4%	10.7%	8.3%	7.0%	8.8%	18.2%
FG	Baseline	40.10±10.10	38.33±12.08	62.21±17.52	59.59±16.47	72.58±22.01	66.62±18.25
	T-CNN	36.63±8.75	33.27±9.68	56.6±20.12	58.67±17.25	64.85±17.58	58.72±16.96
	T-GRU	35.78±8.14	36.52±7.45	58.83±18.17	57.19±18.36	68.2±20.11	61.11±15.55
	TS2P	30.07±6.19	28.98±5.78	49.76±14.12	53.32±13.14	48.11±16.47	43.34±13.32
	Proposed	28.16±7.28	26.06±6.31	45.27±13.33	47.47±14.04	40.27±14.77	37.87±11.71
	Improvement	6.4%	10.1%	9.0%	11.0%	16.3%	12.6%

Since we incorporate the TCN blocks for feature extraction, they are capable of maintaining a much longer receptive field without increasing the number of layers and parameters too much. These extracted features can better represent the variation of the input whole-house load and the corresponding electricity consumption of each individual appliance, which mitigates the over-disaggregated issue that could reduce the disaggregation accuracy.

C. Domain Adaptation Across Datasets

In this part, we evaluate the model performance of the TLN model on the domain adaptation across datasets. We leave one building in a dataset as the target domain without labels, and all the buildings in another dataset are used as the source domain with labels. Three adaptation tasks, $R^A \rightarrow R^3$, $U^A \rightarrow U^2$ and $F^A \rightarrow F^2$, are designed. The performance indices across 100 different testing sample windows are shown in Table IV. As shown, the proposed model performs best among the five methods. A t-test is also applied and the results indicate significant differences between the performance of the proposed model and the others. Note that the improvements of MAE for the washing machine, the dishwasher, the microwave, and the fridge can reach 23.9%, 21.9%, 8.8% and 16.3%, respectively. There are several key reasons for the accuracy improvement of the proposed method: 1) The proposed TCN blocks can extract high nonlinear relationships hidden in these massive load profiles; 2) We use domain adaptation layers to reduce the data distribution discrepancy, and thus the transferable features can be learned; and 3) By jointly optimizing the regression loss and domain adaptation loss, the learned representation can both be prominent thanks to the TCN layers, and unbiased benefiting from the domain adaptation layers.”

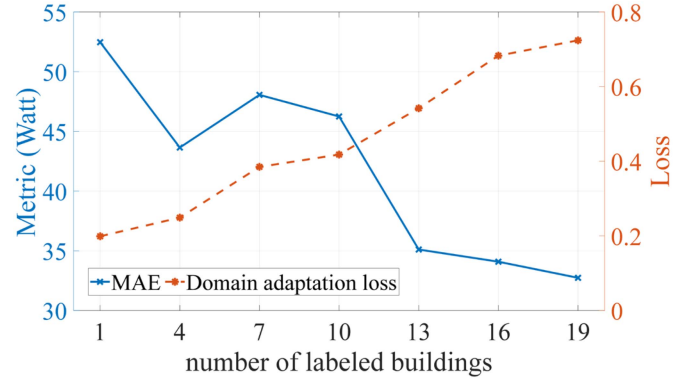


Fig. 8. Comparison of different numbers of labeled buildings from the source domain.

D. Sensitivity Analysis

Effects of the size of datasets: To analyze how the size of the training set affects the domain adaptability and load disaggregation performance, we vary the number of labeled buildings of the source domain while keeping one building as the target domain. Fig. 8 shows the MAE results and the domain adaptation loss for the dishwasher on the task of $F \rightarrow U^2$. Overall, the proposed model trained on larger datasets is more accurate in load disaggregation than that trained on limited data, indicating that sufficient and diverse samples can improve the generalization ability. In addition, it can be seen that when more labeled buildings from the target source are used for model training, the domain adaptation loss will increase since the distribution discrepancy of feature spaces is enlarged. Although there is a certain negative correlation between the MAE and the domain adaption loss based on the evaluation of Pearson’s correlation coefficient, it is not always true that the more labeled data, the

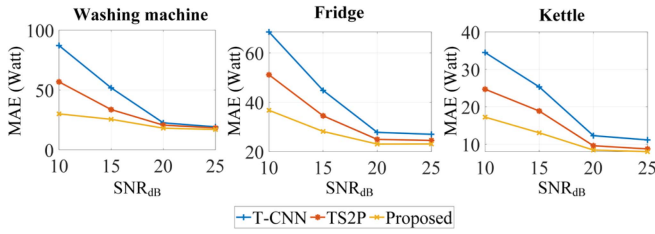


Fig. 9. Comparison of different level of noises.

better performance the proposed method can achieve. The performance of the proposed model trained on 4 labeled buildings is better than that trained on 7 or 10 labeled buildings. This is reasonable as both the regression loss and the domain adaptation loss need to be minimized in the proposed methods. These two losses are equally important in the training process.

Effects of measurement noises: In practical applications, the noises are inevitable when collecting the customers' energy consumption data. Thus, we need to further analyze how the proposed model would perform when the measurements with noises are used. We apply the signal to noise ratio (SNR) to model the measurement noises, which is defined as follows:

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \frac{\mathbf{x}_{\text{real}}}{\mathbf{x}_{\text{noise}}} \quad (17)$$

where $\mathbf{x}_{\text{real}}$ and $\mathbf{x}_{\text{noise}}$ denote the actual values of whole-house load and the added noises, respectively. A larger SNR_{dB} indicates a lower level of noises. When the SNR_{dB} is determined, the $\mathbf{x}_{\text{noise}}$ and the collected measurements with noises $\mathbf{x}_{\text{collect}}$ can be calculated as:

$$\mathbf{x}_{\text{noise}} = \frac{\mathbf{x}_{\text{real}}}{10^{\text{SNR}_{\text{dB}}/10}} \quad (18)$$

$$\mathbf{x}_{\text{collect}} = \mathbf{x}_{\text{real}} + \mathbf{x}_{\text{noise}} \quad (19)$$

In this part, we set $\text{SNR}_{\text{dB}} \in \{10, 15, 20, 25\}$. Each case is run ten times with different random number generator seeds. The average MAE of three selected appliances on the task of $\mathbf{U}^A \rightarrow \mathbf{U}^3$ is shown in Fig. 9. As the value of SNR_{dB} decreases, the level of noises increases, and the MAE results of the three models increase, meaning that there is a decline in the disaggregation accuracy. Nevertheless, the proposed model still performs better than the other methods, and is more robust to the noises, especially for the washing machine whose duty circles are complicated and the fridge that works at very small power consumption.

E. Computational Time and Model Complexity

To evaluate the computational efficiency, we compare the model size and training time of different methods for the kettle in the case of $\mathbf{U}^A \rightarrow \mathbf{U}^2$. The results are presented in Table V. The model size of T-GRU is the smallest, whereas it takes the longest time for model training. It should be mentioned that although it takes some time to train the proposed TLN model, the entire training process is completely offline. Only a few seconds are required to output the disaggregation results given a trained model.

TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY FOR DIFFERENT METHODS ($\mathbf{U}^A \rightarrow \mathbf{U}^2$: KT)

Parameter Description	Model size (Mb)	Training time (Min)
Baseline	4.31	51.05
T-CNN	28.69	43.28
T-GRU	0.75	89.46
TS2P	5.17	38.17
proposed	8.32	48.39

F. Discussions

Applications of NILM results: The proposed model can automatically estimate the energy consumption of individual appliances from the whole-house load. Based on the disaggregated results, we see a series of applications for households and retailers.

For households, the disaggregated results can help them learn more about which appliances account for the largest proportion of the total electricity consumption and how they can better respond to price signals during the peak periods. Moreover, the energy consumption behaviors of the households can be accurately analyzed after obtaining the consumption and operating status of the appliances [33]. More customized home energy management systems can be developed, which are more likely to maximize the benefits and satisfaction of households.

From the perspective of retailers, they can apply the useful information of the disaggregated results on the energy efficiency (EE) and demand response (DR) programs. To improve the effectiveness of EE programs, retailers need to identify a group of customers with higher potential for energy reduction, and it is a critical step to estimate appliance-level energy efficiency in this process [34]. The disaggregated results can help identify the low efficient appliances and the energy-inefficient customers, which will contribute to the development of more effective customer targeted programs. In addition, the knowledge on appliances usage is of great value for DR program implementation [9]. Several tailored DR programs based on the users' consumption patterns as mined from the disaggregated results are proposed in the existing research [8], [9], [35]–[37].

Limitations and Future Work: By minimizing the domain adaptation loss to reduce the distribution discrepancy between the source and target domain, the transferability of the proposed TLN model can be improved, and thus better performance on the unseen buildings can be achieved. Several basic assumptions should be satisfied to guarantee the effectiveness of the TLN model: 1) The learning tasks in the two domains are similar or related. In this paper, the tasks of both domains are disaggregating the energy consumption of individual appliances from the whole-house load; 2) The data distributions between the source and target domains are not too different; and 3) A suitable model can be applied to both domains. Violation of these assumptions may result in negative transfer [38]. For the second assumption, although the domain adaptation loss is computed to measure the data distribution discrepancy in the proposed model, there is no definite scope and basis to determine whether the data distributions are

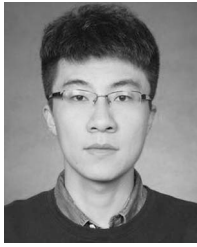
too different. Since the negative transfer is a long-standing and important problem for transfer learning, our future work will focus on investigating how to bound the amount of negative transfer and further improve the model transferability. Another future work will explore the applications of the disaggregated results. One of the directions will be to develop customized DR programs for retailers and end-users.

VI. CONCLUSION

In this paper, a novel methodology with knowledge transferability is developed for NILM in smart grids, which improves prior work particularly respect to accuracy and robustness under scenarios of data shortage, and generalization across domains. The proposed TLN model can fully exploit the source labeled data and target unlabeled data to improve the disaggregation performance of individual appliances energy consumption. The temporal convolutional network with residual connections is developed for dynamic feature extraction. The domain adaptation loss is calculated to reduce the domain distribution discrepancy between the source and target domain representations. Thus, an invariant representation of individual appliances power profile can be learned. The performance and transferability of the proposed method are improved by jointly considering the domain adaptation and energy disaggregation.

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